



# 統計学B

Week 12: 回帰分析 2

Sir Francis Galton demonstrates his “Galton board” or “quincunx” at the Royal

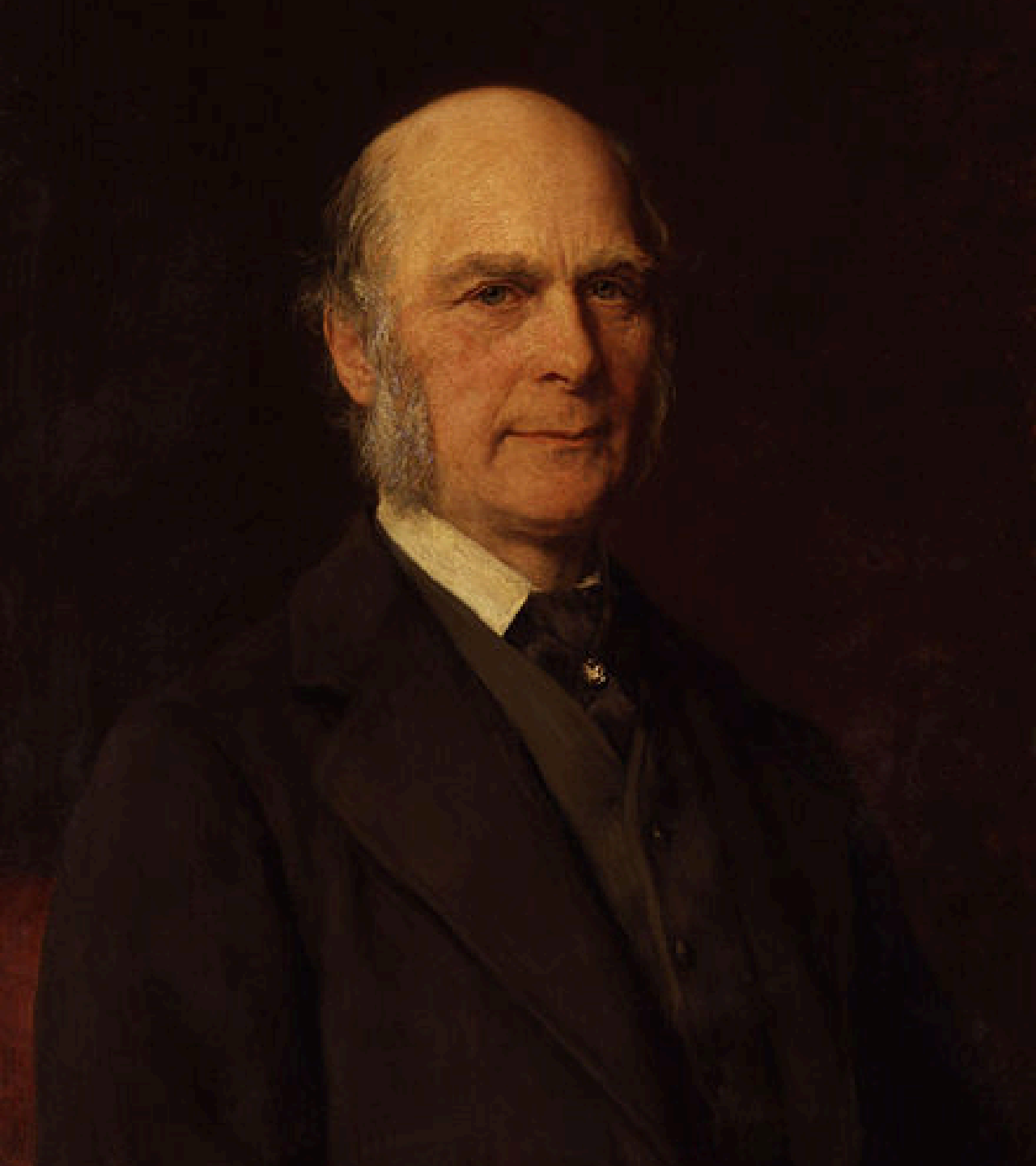
なぜ

「回帰」

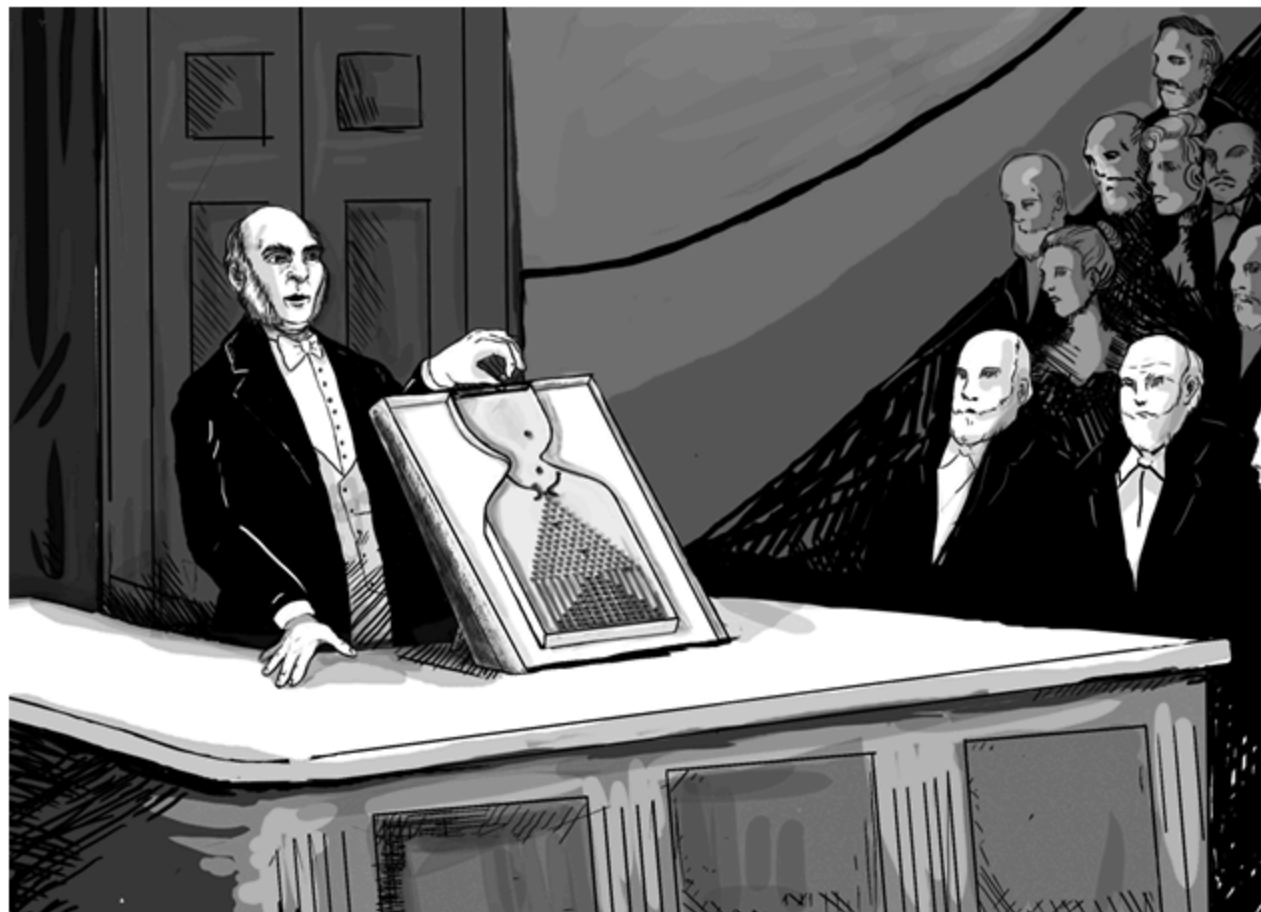
なのか



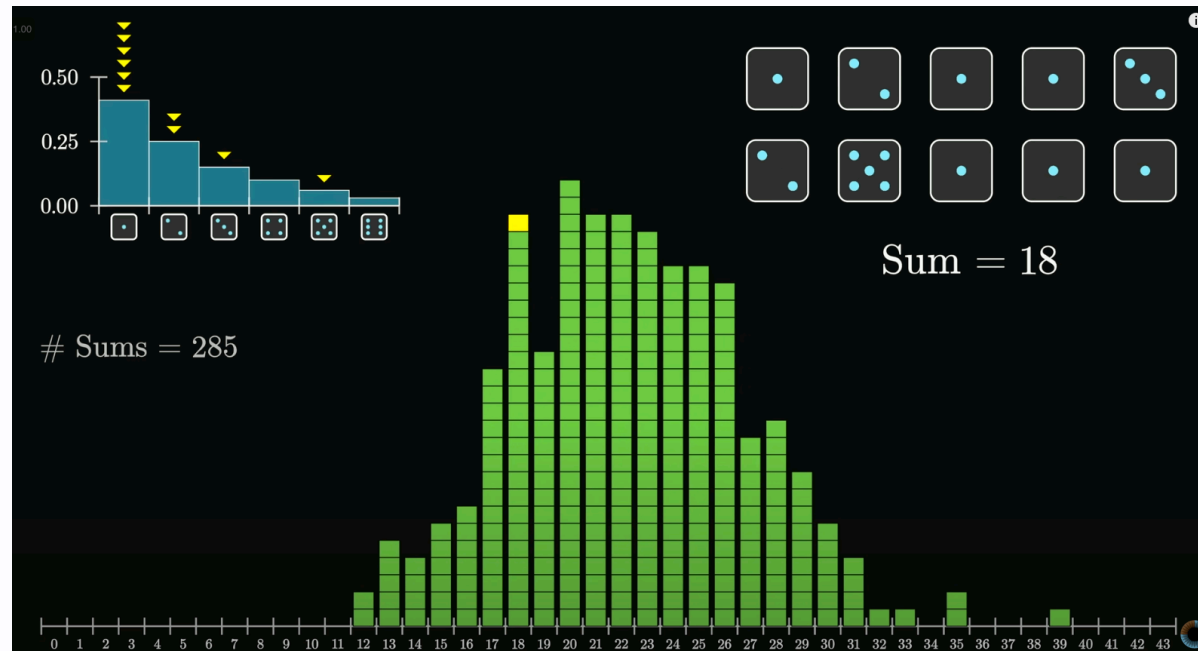
# The Galton Board



- Sir Francis Galton (1822-1911)
- British mathematician
- invented the "Galton Board"
- regression to the mean, or "reversion to mediocrity"
- 回歸先平均  
(ex: test scores, heights, sports)



Sir Francis Galton demonstrates his “Galton board” or “quincunx” at the Royal Institution. He saw this pinball-like apparatus as an analogy for the inheritance of genetic traits like stature. The pinballs accumulate in a bell-shaped curve that is similar to the distribution of human heights. The puzzle of why human heights don’t spread out from one generation to the next, as the balls would, led him to the discovery of “regression to the mean.” (Source: Drawing by Dakota Harr.)



Source: YouTube

# 中心極限定理

Central Limit Theorem

ようにするに...

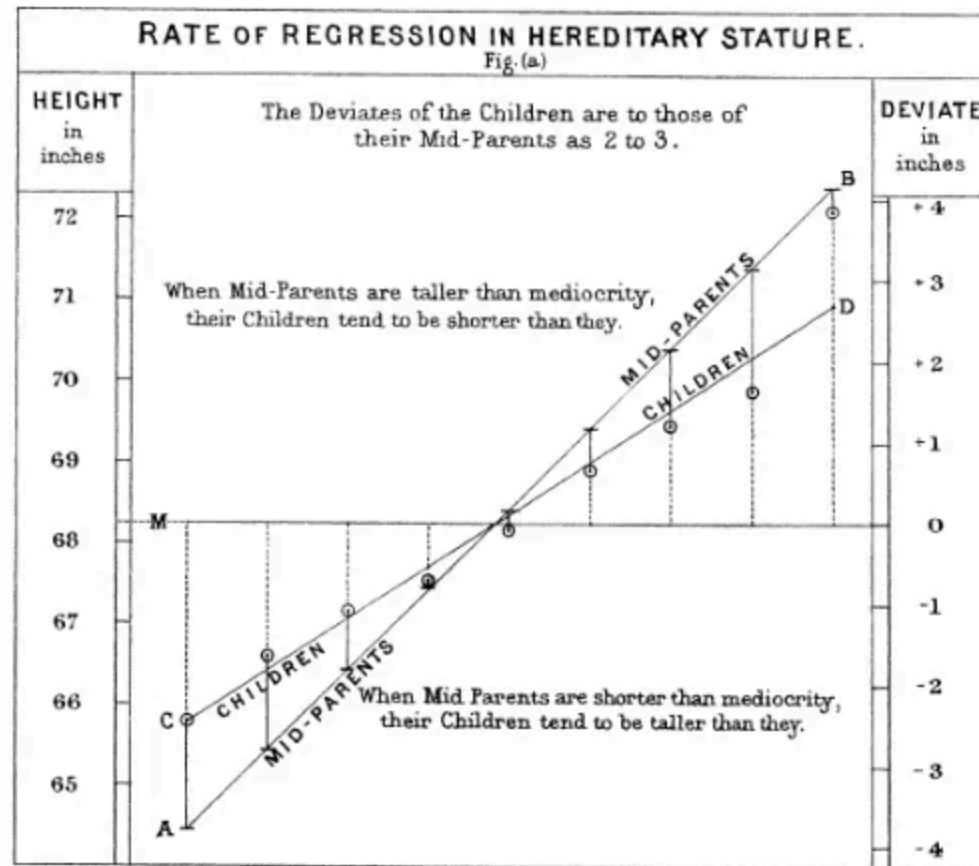
The Central Limit Theorem says that if you take a lot of averages from random samples, those averages will look like a bell curve, no matter how the original data looks!

中心極限定理とは、たくさんのデータをランダムに平均すると、その平均がどんなデータでもベルカーブ（正規分布）に近づくということです！

Galton wanted to see if this idea applied to real life data.

Do tall parents have tall children?  
Do short parents have short children?

Galtonが気づいたのは...  
930人の親子の身長データを使って  
回帰分析を行うと...



回帰分析とは、  
極端な世界が  
平均に引き戻される様子を  
数式で表したもの

極端な値は 平均に戻る

Regression toward the mean  
平均への回帰

だから

「回帰」

なのだ